

Transitive Credit (subsumes previous document)

We take the citation graph to be a graph (V, E) with $E \subset V \times V$. The edge (v_i, v_j) represents the fact that v_i cites v_j . We shall assume that (V, E) is devoid of cycles; it is a d.a.g.

The in- and out-degree of a node n are respectively defined as $\text{indeg}(n) = |\{m \mid (m, n) \in E\}|$ and $\text{outdeg}(n) = |\{m \mid (n, m) \in E\}|$. The normal citation count or *credit* for a node n is simply $\text{indeg}(n)$.

We now consider a model in which credit can be transferred from one node to another – along the edges. Each node gives up some of its incoming credit to its (out-) neighbors. We define the following quantities:

- $\text{tc}(n)$ – the total credit accumulated by a node,
- $\text{ku}(n)$ – the credit kept by a node – its kudos, and
- $\text{tr}(n, m)$ – the credit transferred along the edge (n, m) .

Each node receives the normal citation credit of 1 for each incoming edge. It also receives credit transfer from the nodes that cite it along each such edge:

$$\text{tc}(n) = \sum_{(m,n) \in E} 1 + \text{tr}(m, n)$$

The kudos is the amount of credit a node keeps, which is the total credit less than the amount of credit that flows along the out-edges of that node:

$$\text{ku}(n) = \text{tc}(n) - \sum_{(n,m) \in E} \text{tr}(n, m)$$

Prop. $\sum_{n \in N} \text{ku}(n) = |E|$

This is something that I think we need: that redistributing credit does not create or diminish credit.

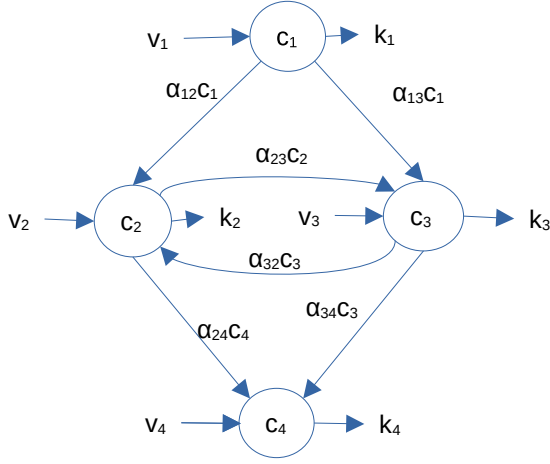
The question is now how to define tr . The simplest method is, I think, for every node to give up a fraction α of its credit keep the remaining $(1 - \alpha)$ for itself a fraction α . It also distributes the outgoing fraction *equally* over the outgoing edges

$$\begin{aligned} \text{tr}(n, m) &= \frac{\alpha}{\text{outdeg}(n)} \text{tc}(n) \\ \text{ku}(n) &= (1 - \alpha) \text{tc}(n) \end{aligned}$$

This has the property that it satisfies the conditions for tc above. In the case that $\alpha = 0$ it is the normal citation count. When $\alpha = 1$ the credit piles up at the terminal nodes of the d.a.g.

Of course, α does not have to be the same for each node, nor does the distribution of credit over the cited nodes need to be uniform, but maybe this is a starting point.

A more general formulation



In this formulation we allow each node to decide how much credit it keeps and how much it gives away. We also allow cycles. Consider node 2 in the picture above. It receives credit from nodes 1 and 3; and rather assuming that it receives one unit of credit for each citation edge, we shall simply assume that it receives an amount v_2 . It *gives* credit to nodes 3 and 4. It takes the total received c_2 , sends $\alpha_{23}c_2$ to node 3, $\alpha_{24}c_2$ to node 4 and keeps the remainder, $(1 - (\alpha_{23} + \alpha_{24}))c_2$ to itself as its kudos, k_2 . So we have

$$c_2 = \alpha_{12}c_1 + \alpha_{32}c_3 + v_2 \quad \text{and} \quad k_2 = c_2 - (\alpha_{23} + \alpha_{24})c_2$$

We can write down all such equations as matrix operations:

$$\begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ \alpha_{12} & 0 & \alpha_{32} & 0 \\ \alpha_{13} & \alpha_{23} & 0 & 0 \\ 0 & \alpha_{24} & \alpha_{34} & 0 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}$$

and

$$\begin{bmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} - \left(\begin{bmatrix} 0 & \alpha_{12} & \alpha_{13} & 0 \\ 0 & 0 & \alpha_{23} & \alpha_{24} \\ 0 & \alpha_{32} & 0 & \alpha_{34} \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right) \times \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}$$

where $+$ and $\times$ are pointwise operations on vectors or matrices of the same dimension. Writing $\mathbf{c}$, $\mathbf{k}$, $\mathbf{v}$ and $\mathbf{1}$ for the credit, kudos, input “citation” and unit vectors; and writing $\mathbf{A}$ for the transfer weights matrix we have:

$$\mathbf{c} = \mathbf{A}^\top \mathbf{c} + \mathbf{v} \tag{1}$$

$$\mathbf{k} = \mathbf{c} - (\mathbf{A}\mathbf{1}) \times \mathbf{c} \tag{2}$$

The *total* credit being input is $\sum_{i=1}^n v_i$ or $\mathbf{1}^\top \mathbf{v}$, which, using 1 and 2 and noting that the inner or “dot” product $\mathbf{x} \cdot \mathbf{y}$ is $\mathbf{1}^\top (\mathbf{x} \times \mathbf{y})$, gives¹

$$\mathbf{1}^\top \mathbf{v} = \mathbf{1}^\top \mathbf{c} - \mathbf{1}^\top \mathbf{A}^\top \mathbf{c} = \mathbf{1}^\top \mathbf{c} - (\mathbf{A}\mathbf{1})^\top \mathbf{c} = \mathbf{1}^\top \mathbf{c} - (\mathbf{A}\mathbf{1}) \cdot \mathbf{c} = \mathbf{1}^\top \mathbf{c} - \mathbf{1}^\top (\mathbf{A}\mathbf{1}) \times \mathbf{c} = \mathbf{1}^\top \mathbf{k}$$

Which is just a sanity check on what goes in must come out and *vice versa* – even when there are multiple solutions for the credit vector $\mathbf{c}$. When do we have multiple solutions? If there are multiple solutions, is there one that is minimal in some sense?

Now, expanding, we have $\mathbf{c} = (\mathbf{I} + \mathbf{A}^\top + (\mathbf{A}^\top)^2 + (\mathbf{A}^\top)^3 + \dots) \mathbf{v}$ and, assuming the series $\mathbf{I} + \mathbf{A}^\top + (\mathbf{A}^\top)^2 + (\mathbf{A}^\top)^3 + \dots$ converges to $\mathbf{W}$, we have $\mathbf{I} + \mathbf{A}^\top \mathbf{W} = \mathbf{W}$.

¹<https://en.wikipedia.org/wiki/Transpose>